

Substrate SM Program — honest-state ledger v0.2 (claim-by-claim derived/assigned/matched scorecard)

Keeper

2026-05-30 Saturday EDT — afternoon consolidation

Honest-state ledger v0.2

Status legend: DERIVED (mechanism proved) · IDENTIFIED (matches <1%, mechanism partial) · MATCHED (value coincides, mechanism not forced) · ASSIGNED (hand-labeled; not yet derived) · FRAMEWORK (structure stated, not closed) · RETRACTED · UNBUILT.

Standing rule (Cal-recurring-pattern, Saturday confirmation): every cell marked DERIVED below is DERIVED modulo the one keystone bet (canonical-basis element of $U_q(B_2)$) IS a physical particle with that K-type). The headline-cap does not drop the conditionality.

L1 — The algebra (engine)

Claim	Tier	Status	Rests on	Flips when
Substrate Hall algebra = $U_q^+(B_2)$ at $q=2$; both Serre relations covered	D	DERIVED / RIGOROUS	E0 (Toy 3597) short-root $[2]_2 = N_c = 3$; E9 (Toy 3610) long-root $[3]_4 = N_c \cdot g = 21$; Ringel's theorem	— (stands; strongest engine result)
Three substrate invariants encoded in the algebra's defining relations (Saturday integration, Lyra+Elie)	D	DERIVED / RIGOROUS	Serre + Drinfeld pairing: $N_c=3$, $N_c \cdot g=21$ (=dim $so(5,2)$), $N_c \cdot n_C=15$ (=dim $Sym^2(5)$)	—
Mersenne-coded structure at $q=2$: $[n]_2 = M_n = 2^n - 1$	D	DERIVED	substrate primaries cluster on low Mersenne ladder ($N_c=M_2$, $g=M_3$, $M_7=127$ anchors N_{max})	— (algebraic side of T2452 Mersenne Tower)

Claim	Tier	Status	Rests on	Flips when
Affine pin: $B_2^{(1)}$ labeling, marks (1,1,2), $\delta = \alpha_0 + \alpha_1 + 2\alpha_2$, $h=4$	D	PINNED (convention)	Lyra affine-pin v0.1 + Keeper refinement	— (note: $B_2^{(1)}$ and $C_2^{(1)}$ are the <i>same</i> untwisted affine for $n=2$ since $B_2 \cong C_2$; labeling chosen to match $SO(5)$ keystone, not selecting between distinct algebras) — (math stands; $\neq$ physics derived)
4/4 Hopf pieces (mult+coproduct, canonical basis, R-matrix, negative part) = Drinfeld double “Goal 1 structurally complete”	D —	RIGOROUS AS ALGEBRA WARRANTED, with caveat	standard quantum-group theory (Lyra engine v0.1-v0.4) the 4 pieces exist	 caveat: “structure complete $\neq$ physics derived” must travel with it

L2 — Kinematics (what particles are)

Claim	Tier	Status	Flips when
5-tuple particle taxonomy — STATIC AXES	partial D / partial ASSIGNED	LEPTON SECTOR DERIVED-modulo-keystone (Lyra #416 v0.1, 18 entries — every static axis: $\sigma_{BF}/region/chirality/charge/particle$ -anti; generation indexed per #414); photon, Higgs single-particle clean; quark + gauge sectors still ASSIGNED pending bulk-color	quark/gauge flip when bulk-color counting layer is dictionarized + Lyra #416 v0.2
3 colors = $h^v(B_2) = N_c = 3$	D	DERIVED (solid)	—
3 generations charge = $SO(2)$ weight (GMN structure located, $T_3 \in SU(2)_L$, Y from $SU(2)_R + SO(2)$)	— D	MATCHED-with-burden DERIVED (mechanism) + LOCATED (coefficients)	see Gates embedding coefficients PINNED-not-asserted, acceptance test $\sin^2\theta_W = 2/9$ (Lyra L2)
Chirality / parity violation = D_{IV}^5 Hermitian $\rightarrow J = SO(2)$ factor $\rightarrow$ holomorphic Hardy $\rightarrow SU(2)_L$ coupling	I	FRAMEWORK-PLUS (Lyra L3)	per-particle L/R labeling pending $SO(5,2) \rightarrow$ Lorentz embedding pin (honestly flagged)

Claim	Tier	Status	Flips when
Region (Shilov/Bulk) = Hardy vs Bergman; lepton-on-Shilov + colorlessness from one fact	D	DERIVED (geometric) modulo keystone	Shilov stabilizer $SO(4)$ carries no color structure (Lyra L2)
Confinement (no Shilov value for non-singlets)	I	DERIVED, FRAMEWORK-PLUS	coupling map multi-week

L3 — Dynamics (the process engine)

Claim	Tier	Status	Flips when
Fusion = Hall product (extension-counting, $GF(2)$); both B_2 Serre relations covered	D	DERIVED AS ALGEBRA (E2/3600 + E9/3610)	physics ID = the dictionary bet
Decay = Green coproduct; β -decay conserves Q/B/L via grading	D	DERIVED AS ALGEBRA (E3/3601)	identification “this coproduct = real β -decay” rides on the dictionary
Conservation = grading	D	DERIVED AS ALGEBRA	engine v0.2 scopes to {Q, B, L} = non-color SM (E8/3609); color extends via bulk-color (#418)
Scattering = R-matrix	—	FRAMEWORK (structure exists; amplitudes not extracted)	Lyra engine v0.3 R-matrix has amplitude extraction multi-week
Absolute mass scale, Higgs/EWSB	—	UNBUILT (least-developed L5)	Lyra multi-week
Per-particle mass derivations (L4 v0.2)	—	UNBUILT (gated on Elie’s bulk K-type radial tower)	Elie bulk-tower + Lyra L4 v0.2
“Goal 2 mechanism built”	—	WARRANTED for the mechanism	the <i>mechanism</i> is real; <i>physical model</i> needs the dictionary

L4 — Composites / nuclear

Claim	Tier	Status	Flips when
Proton = bulk $k=6$ ($C_2=6$) = YM mass gap; $m_p = 6\pi^5 \cdot m_e$	D	DERIVED (0.01%)	— (anchor)
m_p/m_e structurally organized: $6\pi^5 = C_2 \cdot \pi^{(n_C)}$ (adjoint Casimir $\times$ Bergman volume)	I	FRAMEWORK-PLUS (Lyra L4 v0.1 reading)	dictionary closes the mass-mechanism story

Claim	Tier	Status	Flips when
Nuclear shells = $SO(5)$ K-type closures ($SO(5)$ = compact factor of D_{IV}^5)	I	STRUCTURAL / HYPOTHESIS	$SO(5)$ shell-closure computation (Elie E11): does branching give 2,8,20,28,50,82,126?
Magic-number forms (rank-products); 126 = Universal-Q; spin-orbit = C_2/n_C	S	SUGGESTIVE (Grace-flagged)	E11 derivation; small-target/coincidence-denominator caveat until then

L5 — Engineering

SP-30 falsifier designs: EXIST. Control theory: UNBUILT. (Same open dynamics layer as L3.)

L6 — Periodic Table

Claim	Tier	Status	Flips when
v0.4 structure (Grace): 3 generation rows $\times$ 4 family columns + boson block + composite layer (Elie E10 backbone, 10 cells $\dim \leq 35$)	—	BUILT	v0.5 with E10 absorption Saturday
Fundamental-sector selection (4-backbone) = {trivial + ω_1 + ω_2 + adjoint} of $B_2=so(5)$	FRAMEWORK-PLUS	rep-theory part RIGOROUS (fundamental weights/adjoint); $so(5)$ -SPECIFIC strengthened: matter = 4-dim Dirac spinor (not a generic Lie fact); σ_{BF} matter/force split clean	DERIVED when dictionary closes per-sector identifications
Lepton row per-particle	D	DERIVED modulo keystone bet (Lyra #416 v0.1, first row flipped)	—
Quark + gauge rows per-particle	—	ASSIGNED	bulk-color counting layer + Lyra #416 v0.2
Gaps = predictions (no 4th row, no GUT/SUSY/steriles)	—	MATCHED $\rightarrow$ prediction	row-count forcing closes the no-4th-row gap

The gates (open work)

Gate	Status	Detail
A_sub↔Hall dictionary (the keystone)	PHASE 2 FIRST WAVE LANDED	L1 keystone laid; L2 Region+GMN derived; L3 chirality FRAMEWORK-PLUS; L4 mass framework + $\alpha=1/N_{\max}$ placed (L8) + Macdonald-Koornwinder placed (L9); lepton row per-particle DERIVED. Per-particle quark/gauge waits on bulk-color counting layer. K2 K-audit CONDITIONAL PASS, 3 minor-moderate findings — Lyra v0.2 of L1+L2 will absorb. (Lyra #412+#414 v0.2). Count-NUMBER over-determined/forced as a B_2 invariant ($h^\vee=N_c=3$); generation-IDENTIFICATION mechanism OPEN with the burden: produce TWO <i>independent</i> 3-fold structures from one invariant. Route (I) Coxeter/ δ UNDERCUT; route (II) Track P favored. Earlier “two independent 3’s” framing SUPERSEDED → “one 3 ($h^\vee$) double-duty.” #414 v0.2 + Saturday integration: the two 3-folds are addressed structurally via bulk-color v0.2 (color = $SO(3)$ sub-vector, geometric) + spinor ³ -channel candidate (E7) for generations (algebraic).
Generation-forcing	TENSION RELIEVED, NOT CLOSED	
Bulk-color mechanism — TWO SUB-GATES (Cal Saturday refinement)	see below	The bulk-color story is naturally two layers. The counting layer can promote independently of the gauge-phenomenology layer.

Gate	Status	Detail
(a) Counting layer — 3 substrate directions for color	ADDRESSED structurally	Lyra v0.2 SO(3) sub-vector candidate: $5(\text{SO}(5)) \downarrow \text{SO}(3) \times \text{SO}(2) = (3,0) + (1,\pm 1)$ — the 3-dim SO(3) sub-vector inside the bulk K-type's SO(5)-vector content. Elie A1 (Toy 3612) supplies the rigorous Lie-theoretic backbone: explicit $\mathfrak{so}(5,2) = \mathfrak{k} \oplus \mathfrak{p}$ Cartan decomposition, K-rep of $\mathfrak{p} \cong V_{\text{so5}(\text{vec})} \otimes V_{\text{so2}(\pm)} =$ doubled-vector-with-charge. Two routes for direct SU(3) subgroup CLOSED (SU(3) not in K; $[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$ so not in $\mathfrak{p}$ either). Promotion candidate: rep-theory RIGOROUS, candidate FRAMEWORK-PLUS, modulo keystone.
(b) Gauge-phenomenology layer — SU(3) QCD emergence	OPEN, multi-week	Must derive: confinement, asymptotic freedom, 8 gluons, running coupling — from substrate SO(3) 3-direction counting (Track P / counting-not-symmetry). Two narrowed routes per Elie A1: (C) hidden dynamics-symmetry on Hardy-space sections built from p-operators, OR (D) pure counting without literal SU(3) algebra. Lyra/Grace family-map confirms Family (1) Lie-subgroup-on- $\mathfrak{p}_+$ + candidates excluded (Elie $[\mathfrak{p}_+, \mathfrak{p}_+] = 0$ obstruction).
α -placement	PLACED (Lyra L8)	$\alpha = 1/N_{\text{max}} = 1/137$; $\alpha^{-1} = 2^{\wedge}g + N_{\text{c}}^2 = 128 + 9 = 137$ (integer leading order; $0.036 \approx \text{QED loops}$). Cleanest derived-constant entry of the week.
Baryogenesis α^4 in $\eta = 2\alpha^4/(3\pi)$	LOCATED, not sized	E $\leftrightarrow$ F asymmetry (Lyra L7); sizing requires α^4 derivation from CP-violating amplitude minimum order (open, #399 multi-week)
A1 PRIMARY paper final-PASS	OVERDUE CAPSTONE	Cal-flagged Friday: “one Keeper read away since Thursday.” Residual = FK genus book-pin. Saturday afternoon Lyra v0.4 sweep + Keeper PASS + Cal cold-read would close to submission-grade.

Gate	Status	Detail
C12-C14 T-number/META audit	OVERDUE (Cal #156 Flags A+B)	T2467/T2468/T2469 → C12/C13/C14 mislabels (C13 “SCMP(T2468)” wrong — SCMP is T2469); Cal #99 META distinction on T2467 promotion. Gates external citation of Strong-Uniqueness v1.1 across paper series. Joint Lyra+Keeper.

Standing predictions (falsifiable)

- ν mass ordering = NORMAL (conditional on shared-W_n) — JUNO/DUNE 2025-2030.
- Dirac neutrinos, $m_{\beta\beta} = 0$, no $0\nu\beta\beta$ at any scale — LEGEND-1000/nEXO.
- PMNS angles — $\sin^2\theta_{12} = \text{rank} \cdot N_c \cdot g / N_{\text{max}} = 42/137$, $\sin^2\theta_{23} = N_c \cdot n_C^2 / N_{\text{max}} = 75/137$, $\sin^2\theta_{13} = N_c / N_{\text{max}} = 3/137$. 3/3 within 1σ , substrate-primary numerators. F1 falsifier (Lyra-sharpened Saturday): if any angle pins outside substrate-primary form, framework refuted on the mixing-angle spine.
- K-types-are-particles falsifier T1 (Cal proposal, joint design w/ Lyra) — dim of canonical basis at $V_{(1/2,1/2)}$ compared to a-priori-pinned SM-observable count; forced YES/NO via standard QG computation, no dictionary-bet leakage.
- No 4th generation, no GUT, no SUSY, no steriles (table closure / Five-Absence).
- $\hat{B}_2$ tube count — stakes lowered (route II carries the count regardless); if Elie’s #409 recompute returns $\neq 3$ it’s informative not decisive.

Consistency fixes (committed ee35f15 Friday)

- gravity = S^1 /EM boundary condition, not a force (data/bst_forces.json).
- neutrino character = Dirac (Paper 108 corrected).

Methodology guard

- Source-Verification Tier — Calibration #33 STANDING (Cal ratified Saturday AM via Methodology Index v0.9 addendum 2, Q15). Four-level discipline stack now COMPLETE: Cal #27 result / #29 design / #32 slot / #33 sourcing. RECALLED capped at MATCHED on a load-bearing literature value; scope is part of the value.
- Headline-cap-conditionality pattern (Cal recurring observation, Saturday): “DERIVED-modulo-explicit-bet must travel with the bet flag; headline-cap doesn’t get to drop the conditionality.” Recurring 3× now (Cal #155 + #32 resolution protocol + K2 Finding 3). Pattern-watch toward Calibration #34.
- Honest-negative-strengthens — 10th instance recorded Saturday (Cal #165 on Lyra #414 v0.2). Standing strength of the audit chain.

K-audits filed this session

- K1 Engine Consolidation v0.1 — verdict CONDITIONAL PASS, 3 conditions (Keeper_KAudit_Dynamics_Engine_Consolidation v0.1 Pre-audit memo for v0.2 filed (Keeper_PreAudit_Engine_v0_2_for_Elie.md): all 3 conditions substantively resolved by Elie’s E8/E9 toys + Lyra #414 v0.2; Elie’s v0.2 doc consolidation in progress Saturday AM (he self-caught the “claimed-done-actually-pending” pattern within minutes).
- K2 Dictionary First Wave — verdict CONDITIONAL PASS, 3 findings (Keeper_KAudit_Dictionary_FirstWave_v0_1.md). Covers L1 + L2 + L3 + L4 framework + #414 v0.2 + #416 lepton flip. Lyra v0.2 of L1+L2 will absorb.

The honest one-line verdict

The engine is real and rigorous as algebra (4/4 Hopf, both Serre relations covered, Mersenne-coded, $B_2^{(1)}$ pinned); the dictionary's first wave landed (L1-L4, lepton row flipped per-particle, chirality mechanism derived, α placed at $N_{\max}$); the bulk-color counting layer has a structural candidate ($SO(3)$ sub-vector inside the bulk K-rep); proton/charge/confinement/mixing-spine + PMNS 3/3 within 1σ are genuinely DERIVED-modulo-keystone-bet; the deepest gate is honestly mapped (count forced, mechanism open-with-burden, two-structures burden structurally addressed via bulk-color v0.2 + spinor³ candidate). What remains UNBUILT: $SU(3)$ gauge-phenomenology emergence (multi-week), explicit per-particle masses (L4 v0.2, gated on Elie's bulk K-type tower), L5 absolute scale, $SO(5)$ shell-closure derivation, and the two overdue capstones — A1 PRIMARY paper final-PASS + C12-C14 T-number audit. Strong, honest place. Not finished.

— Keeper, honest-state ledger v0.2, 2026-05-30 Saturday afternoon. Updates as: Elie engine v0.2 lands (re-audit → clean PASS); Lyra L1+L2 v0.2 absorbs K2 findings; bulk-color gauge-phenomenology layer surfaces a derivation; A1 PASS closes the overdue capstone; #409 recompute settles or doesn't.